



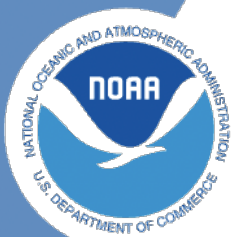
NOAA Technical Memorandum NMFS-XXX-##

Makara Data Submission Guide

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Genevieve Davis

January 2023

U.S. DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



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Makara Data Submission Guide

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Table of contents

Executive Summary	1
1 General_Information	2
1.1 General Information	2
2 2_Document_History_and_Revisions	3
2.1 Document History and Revisions	3
3 Contact_Information_for_the_Makara_Database_Support_Team	5
3.1 Contact Information for the Makara Database Support Team	5
4 4_Tips_and_Recommendations	6
4.1 Tips and Recommendations	6
5 5_Data_Table_Template_Instructions	7
5.1 Data Table Template Instructions	7
6 6_Primary_Data_Tables	8
6.1 Primary Data Tables	8
7 7_Optional_Data_Tables	10
7.1 Optional Data Tables (Recommended/Required if Applicable)	10
8 8_Reference_Tables_Maintained_by_NEFSC	13
8.1 Reference Tables Maintained by NEFSC	13
9 9_Data_Type_Formatting_Instructions	16
9.1 Data Type Formatting Instructions	16
10 10_Data_Table_Field_Definitions	21
10.1 Data Table Field Definitions	21

Table of contents

11	11_Deployments_Table_Field_Definitions	23
11.1	Deployments Table Field Definitions	23
12	12_Recordings_Table_Field_Definitions	25
13	13_Analyses_Table_Field_Definitions	26
14	14_Detections_Table_Field_Definitions	27
15	15_Devices_Table_Field_Definitions	28
16	16_Projects_Table_Field_Definitions	29
17	17_Sites_Table_Field_Definitions	30
18	18_Recording_Intervals_Table_Field_Definitions	31
19	19_Tracks_Table_Field_Definitions	32
20	20_Track_Positions_Table_Field_Definitions	33
21	21_Sensor_Dataset_Field_Definitions	34
22	22_Sensor_Values_Table_Field_Definitions	35
23	23_Frequently_Asked_Questions	36
24	24_Example_Scenarios	37

List of Figures

- 6.1 Diagram showing how the primary tables are related. Each organization owns one or more deployments, for which there can be one or more recordings. The recordings are then analyzed, generating detections. . 9

List of Tables

Executive Summary

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1 General_Information

1.1 General Information

This guide provides data contributors with instructions, field definitions, and examples for how to enter acoustic data and metadata into the template tables and submit them for archiving in the Northeast Fisheries Science Center's (NEFSC) Makara Database. Data that are archived in Makara may also be displayed on the Passive Acoustic Cetacean Map (PACM), with permission from the data owner(s).

2 2_Document_History_and_Revisions

2.1 Document History and Revisions

Ver- sion	Date	Author	Description
1.0	May 1, 2025	Juan Lianne Wilder	Document Created
1.1	June 10, 2025	Jefrey Walker	Updated "Data Table Template Instructions" to clarify that filenames and folder structure must match "templates" sub-directory.
1.2	August 26, 2025	Jefrey Walker	- Renamed the organization_code to deployment_organization_code in the analyses and detections tables to clarify that this field refers to the organization that owns the deployment. Also added analysis_organization_code to the detections table. Both tables now include both the deployment_organization_code (organization that the deployment and recording(s) belong to), and analysis_organization_code (organization that the analysis belongs to). - Added track_uri to tracks table for linking tracks to external files in cloud storage (useful for linking track metadata to high resolution data not stored in makara) - Fixed minor typos

2 2_Document_History_and_Revisions

1.3	October 7, 2025	Jefrey Walker	• Clarified that deployments, recordings, analyses and detections are the primary data tables, but not always required for each submission • Added Sensor Datasets and Sensor Values templates • Added deployment_realtime and deployment_url fields • Updated JSON guidelines and examples • Set default value for recording_n_channels to 1 • Fixed analysis_processing_types to have values of POST_PROCESSED and REAL_TIME (underscores instead of dashes) • Clarified descriptions of analysis_release_data and analysis_release_pacm
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3 Contact_Information_for_the_Makara_- Database_Support_Team

3.1 Contact Information for the Makara Database Support Team

If you have questions or issues regarding your data submission, please contact nmfs.nec.pam-data@noaa.gov.

4 4_Tips_and_Recommendations

4.1 Tips and Recommendations

- Timezones/Timestamps: All timestamps that are entered into the data submission templates should include the timezone (even if they are in UTC; e.g., 2023-10-12T18:09:05+0000). The NEFSC generally recommends that all recordings are made in UTC. Other timezones are okay, so long as the timezone information is included within the timestamp (e.g., 2023-10-12T18:09:05+0500 to indicate that the audio was recorded in Eastern Standard Time).
- Scenarios and Examples: It is highly recommended that data contributors look through our Scenarios section and the 'examples' template folder in the downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage prior to entering their data into the data submission templates. There may be an example that is similar to their situation.

5 5_Data_Table_Template_Instructions

5.1 Data Table Template Instructions

To submit data and metadata to the Makara database, the information must be organized into template tables. The data template tables and example tables are provided in a downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage, and are described in more detail below. Some of the tables are required, but do not necessarily need to be submitted at the same time. Other tables are optional, but it is highly recommended that the data contributor fills out any tables that are applicable to their deployment(s). Within each table, there are required and optional (but recommended if applicable) fields. The required tables/fields are the bare minimum of what metadata should be submitted.

Within the template zipped folder is a sub-directory named "templates/". This sub-directory contains empty CSV files that can be used as starting templates for filling out each table. Each data submission should contain one or more of these CSV files with data rows properly entered based on the formats described below. The names of each CSV file as well as the folder structure must match what is in the templates folder. Data cannot be submitted as an Excel Workbook (*.xlsx file).

6 6_Primary_Data_Tables

6.1 Primary Data Tables

The following tables are the primary tables for any data submission. Together, these four tables represent the core data stored in Makara. However, they are not strictly required for every submission. For example, a submission could contain only deployments but no recordings, analyses or detections, which could be added in a later submission. For each table, the data only needs to be submitted once unless updates or corrections are needed. In those cases, a revised table may be submitted so long as there are no changes in the primary codes identifying each row (e.g., deployment_code, recording_code, etc.).

Data Table	Definition	When to Submit
Deployments	The use of a platform (e.g., bottom-mounted mooring, buoy, glider, towed array) to record audio over a period of time	After a deployment has ended (the start date/time of a deployment is a required field, so a partial table could be submitted once a recorder is in the water, but would need to be updated after retrieval)
Recordings	Single or multichannel recordings that were made over the course of one or more deployments	After a deployment has ended
Analyses	The metadata needed to understand the detection results (i.e., the parameters of the analysis protocol)	After the data have been analyzed (if applicable, at the same time as the Detections table)

Detections	A time-binned or event-based identification of a sound source (e.g., dolphin, Atlantic cod, humpback whale, rain, vessel) produced from a passive acoustic detector or manual analysis that is used to determine presence of the sound source	After a recording has been analyzed (if applicable)
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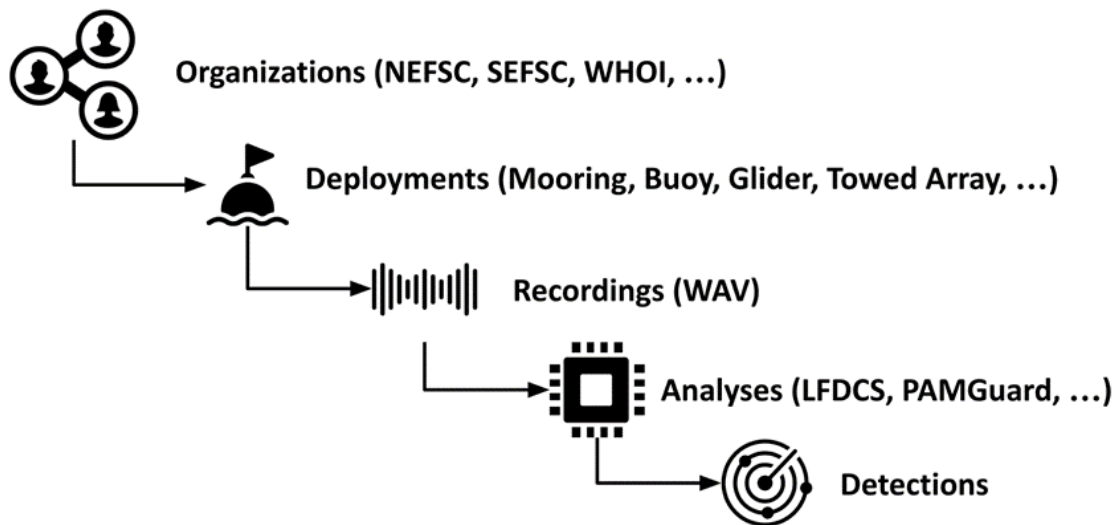


Figure 6.1: Diagram showing how the primary tables are related. Each organization owns one or more deployments, for which there can be one or more recordings. The recordings are then analyzed, generating detections.

7 7_Optional_Data_Tables

7.1 Optional Data Tables (Recommended/Required if Applicable)

The following tables are optional for any data submission as long as the desired categories already exist in the database. For each table, the data needs to be submitted only once. For example, if a device has already been submitted with a deployment, and then referenced by a deployment that is submitted later, then the Device table including that device does not need to be included in the second submission as it will already exist in Makara.

Data Table	Definition	If/When to Submit
Devices	A specific piece of equipment used as part of a deployment (e.g., platforms, hydrophones, recorders, sensors, acoustic releases, gliders, arrays)	If there are new devices that are not already in Makara

7 7_Optional_Data_Tables

Projects	A group of deployments	If multiple deployments are submitted that the data contributor would like to have grouped together, or if new deployments are submitted that should be grouped with deployments that are already in Makara. If a project was already created with previously submitted deployment(s) and there are new deployments to be added to that project, then another Project table does not need to be submitted.
Sites	The fixed location(s) where recordings were made over the course of one or more deployments, or general region for mobile deployments	If there are new deployment sites that are not already in Makara
Recording Intervals	The range of time and/or frequencies with compromised or invalid audio (this table is not for tracking duty cycled data)	If a deployment has periods of compromised data (duty cycling is accounted for in the Recordings table, not the Recording Intervals table)
Sensor Datasets	Metadata about the collection of sensor readings (e.g., temperatures)	If sensor data (e.g., temperatures) were collected on the deployment
Sensor Values	The timestamped sensor readings belonging to a Sensor Dataset	If sensor data (e.g., temperatures) were collected on the deployment

7 7_Optional_Data_Tables

Tracks	Metadata about the collection of track data (e.g., position, speed, and other variables) associated with a mobile platform	If a mobile deployment is submitted
Track Positions	The instantaneous timestamped location, speed, etc. along the track of a mobile platform	If a mobile deployment is submitted

8 8_Reference_Tables_Maintained_by_-NEFSC

8.1 Reference Tables Maintained by NEFSC

NEFSC maintains the reference tables that are listed below. An updated copy of each table can be accessed by downloading the Excel document on the Passive Acoustic Reporting System Templates webpage. **If a data contributor is submitting data, but does not see the correct or applicable category that they need in one of these tables, they can contact the Makara Database Support Team to request that the category be added.** For example, if the submission includes detection data from an acoustic detector that does not already exist on the Detectors table in Makara, then the data contributor can contact the support team and have the new detector added to the reference table.

Reference Table	Definition
Analysis Granularity Types	The granularity of an analysis (e.g., time-binned intervals, call-by-call events, encounter events)
Analysis Processing Types	The processing method of an analysis (e.g., real-time, post-processed)
Analysis Quality Types	The analysis quality indicating the level of detection validation (e.g., full, partially, or not validated)
Behaviors	The behavior associated with a detection event (e.g., foraging, socializing)
Call Types	The unique call/click types associated with one or more sound sources/species (e.g., sperm whale clicks, North Atlantic right whale upcalls)

8 8_Reference_Tables_Maintained_by_NEFSC

Call Type / Sound Sources	A list of call type codes and sound source codes indicating which call types can be used for which sound sources. This table can be used to find the call types available for a given species by filtering for that species sound_source_code.
Demographics	The demographics associated with a detection event (e.g., adult male, female or juvenile)
Detection Quality Types	The quality of detection type indicating whether it was validated or not (e.g., unverified, valid)
Detection Result Types	The detection result types based on the analysis methods (e.g., detected, possibly detected, not detected)
Detectors	The acoustic detector software programs used to process raw acoustic data (e.g., LFDSC, PAMGuard's whistle and moan detector)
Device Types	The general types of devices used as part of a deployment (e.g., platforms, hydrophones, recorders, sensors, acoustic releases, gliders, arrays)
Localization Methods	The methods used to localize the position of individual sound sources (e.g., DMON real-time tracking, time difference of arrival)
Localization Depth Methods	The methods used to localize the depth of individual sound sources (e.g., 3D Simplex, automatic direct reflected time difference of arrival)
Organizations	Any organizations that are registered in Makara (e.g., NEFSC, Woods Hole Oceanographic Institution, Cornell University)
Platform Types	The types of platforms upon which the hydrophones are mounted (e.g., gliders, moorings)

8 8_Reference_Tables_Maintained_by_NEFSC

Recording Quality Types	The recording quality types indicate whether a recording is good, compromised, unusable, etc.
Sensor Variables	The variables that are measured by a sensor (e.g., temperature)
Sound Sources	Specific biological, anthropogenic, environmental, or unknown sound sources (e.g., North Atlantic right whale, recreational vessel, rain)

9 9_Data_Type_Formatting_Instructions

9.1 Data Type Formatting Instructions

Some fields on the data table templates require specific formatting when entering values.

Data Type	Required Formatting	Examples
Boolean	TRUE or FALSE	TRUE
Code	Uppercase letters, numbers, underscores (_), and dashes (-) only (no spaces, lowercase letters, or other symbols)	"NEFSC_MA-RI_-202202_COX01",
		"SOUNDTRAP-53423"
Code List	Comma-separated list of Codes within a single string	"SOUNDTRAP-53423,VEMCO-9867,FPOD-12355"
Date	Date as "YYYY-MM-DD"	"2025-03-26"
Float	Decimal number	72.56281
Integer	Integer number	1
JSON	Valid JavaScript Object Notation (JSON) for storing structured data objects and arrays (see JSON Instructions)	{ "KEY": "VALUE", "KEY": [1, 2, 3, ...], ... }
Text	Text string (no character limit)	"Recorder lost at sea"

Times	Date and time in one of the following ISO-8601 formats	
tamp	(seconds and milliseconds are optional), must include timezone suffix (use 'Z' for UTC):	
	"YYYY-MM-DDTHH:MMZ"	"2025-03-26T15:34Z"
	"YYYY-MM-DDTHH:MM:SS±HH"	"2025-03-26T15:34:27+04"
	"YYYY-MM-DDTHH:MM:SS±HHMM"	"2025-03-26T15:34:27-0500"
	"YYYY-MM-DDTHH:MM:SS.SSS±HHMM"	"2025-03-26T15:34:27.921-0500"
Time-	Timezone as UTC offset in one of the following formats:	
zone	"UTC"	"UTC"
	"UTC±H"	"UTC+4"
	"UTC±HH"	"UTC+04"
	"UTC±HHMM"	"UTC-0500"
URI	Valid Uniform Resource Indicator (URI) address for files in cloud storage. Must start with the protocol (e.g., "gs://" for Google Cloud buckets), followed by the bucket name, and then path to the file within the bucket.	"gs://nefsc-1/bottom_-mounted/NEFSC_-GOM/NEFSC_GOM_-200410/NEFSC_-GOM_200410_-PU05"
URL	Valid Uniform Resource Locator (URL) address for web page	"https://dcs.who.edu/gom2412/gom24we14.shtml"

9.1.1 JavaScript Object Notation (JSON) Instructions

JSON (JavaScript Object Notation) is a lightweight data interchange format that is easy for humans to read and write and for machines to parse and generate. In Makara, several fields are designated as JSON type to store structured data that doesn't fit well into standard columns. For example, information about device calibration, channel maps, and detector settings can all be stored as JSON for loading directly into other software.

9.1.2 When to Use JSON Fields

The JSON fields are primarily intended to be used in the following situations:

- To store structured data that benefits from consistent data structures (e.g., calibration tables)
- When you need the data to be directly queryable within the database
- When the data can be read or processed directly by other software applications

JSON is particularly valuable for device specifications, channel mappings, calibration data, and complex configuration settings. However, if your additional information is primarily free-text notes or simple comments that don't require structured querying or programmatic access, NEFSC recommends using the standard comments field available in each table instead. The comments fields are simpler to fill out, don't require special formatting, and are better suited for human-readable annotations that don't need computational processing.

9.1.3 What is JSON?

JSON consists of two primary structures:

- **Objects:** Collections of key-value pairs enclosed in curly braces { }. Each key must be a string in double quotes, followed by a colon :, and then a value. Example: { "DETECTOR_THRESHOLD": 0.75 }
- **Arrays:** Ordered lists of values enclosed in square brackets []. Example: [0.5, 0.75, 0.9]

Values in JSON can be:

- **Strings:** Text in double quotes "example"
- **Numbers:** Integers or decimals without quotes 42 or 3.14
- **Booleans:** true or false (lowercase, without quotes)
- **null:** Represented as null (without quotes)

There are no specific formats required for dates and timestamps. However, NEFSC recommends using the same formats listed in the Data Type Formatting Instructions table based on the ISO8601 standard (e.g., { "date": "2025-03-20", "datetime": "2025-03-26T15:34:27Z" }).

9.1.4 Example JSON Structures

Simple object for the sensitivity of a recorder:

```
{ "SENSITIVITY": -1.8 }
```

Nested object for the high and low gain sensitivities of a recorder:

```
{ "GAIN": { "HIGH": 177.4, "LOW": 189.9 } }
```

Object with nested array for the sensitivity calibration of a hydrophone:

```
{ "SENSITIVITIES": [-170.1, -169.8, -170.2], "CALIBRATION_DATE": "2023-05-15" }
```

Complex nested structure containing a calibration frequency table represented as an array of objects where each object within the array is a row in the table:

```
{  
  "FREQ_TABLE": [  
    { "FREQ_HZ": 1, "GAIN_DB": 10.6 },  
    { "FREQ_HZ": 2, "GAIN_DB": 20.3 },  
    { "FREQ_HZ": 5, "GAIN_DB": 35.8 },  
    { "FREQ_HZ": 10, "GAIN_DB": 47.9 },  
    etc...  
  ]  
}
```

9.1.5 Recommended Guidelines

NEFSC recommends adhering to the following guidelines when creating JSON data for submission to Makara. Please note these are only guidelines, not requirements. The only requirement for these fields is that the data is in valid JSON format.

- Each entry in a JSON field should be an object at the top level, not an array or individual value so that all values have a key specifying what they are (e.g., use `{"SENSITIVITIES": [-170.1, -169.8, -170.2]}` instead of just the array of values `[-170.1, -169.8, -170.2]`)
- Always use uppercase letters for object keys to avoid changes in case across rows (e.g., `{"HIGH": 177.4}` and `{"high": 177.4}` are not compatible)
- Do not wrap numeric values in double quotes, leave them as numbers (e.g., `{"HIGH": 177.4}` instead of `{"HIGH": "177.4"}`)

9.1.6 Tips for Working with JSON in Spreadsheets

- Create your JSON in a text editor or JSON validator first to ensure proper formatting
- For complex JSON structures, consider using a JSON validation tool (like jsonlint.com) before pasting into your spreadsheet
- Keep JSON structures as simple as possible while still capturing the necessary data

9.1.7 Consistency is Critical

It is the responsibility of the data contributor to adhere to consistent structures (aka schema) within the JSON field of each table. This is critical for ensuring data integrity and usability, and so JSON data can be readily combined across multiple rows in a given table. While NEFSC provides some example formats for each JSON field in this guide, each data contributor can develop their own JSON schema that aligns with their software and workflow.

10 10_Data_Table_Field_Definitions

10.1 Data Table Field Definitions

The following tables contain definitions for each field or column in the data table templates contained in the zipped folder. Each of the following field definitions tables in this document has a number of columns which are defined below in the Column Header Key section, including columns for specifying which fields are required, the type of data that should be entered (e.g., text, integer, URL), and which reference tables to refer to when filling out that field.

10.1.1 Column Header Key

- **Field:** The name of the field
- **Definition:** The definition of the field with examples of formatting and what to include
- **Required:** Indicates whether the field is required (i.e., 'Always') or conditional (e.g., "If recording_device_lost=FALSE, then this field is always required"). Some fields are required if the data are also being submitted to the National Centers for Environmental Information (NCEI). If those fields are not already required for submitting data to the Makara database, then there will be a note in the Required column saying "Required for NCEI".
- **Default:** Indicates whether the field has a default value if no other value or text is entered
- **Data Type:** The data type accepted by that field (see Data Type Formatting Instructions)
- **Reference Table:** The reference table that contains the category options for each field. Some reference tables are labeled with "(see Excel)", indicating that they are global reference tables available in the "Makara - Reference Tables" Excel document* in the zipped folder. If a reference table is listed here but not found

10 10_Data_Table_Field_Definitions

with the rest of the reference tables, it is likely a reference table that is maintained internally by NEFSC and the categories can be searched for directly in Makara.

- **Constraint:** Any constraints on the contents of that field

**Note: An updated copy of each reference table can be accessed by downloading the Excel document on the Passive Acoustic Reporting System Templates webpage. If a data contributor is submitting data, but does not see the correct or applicable category that they need in one of these tables, they can contact the Makara Database Support Team to request that the category be added.*

11 11_Deployments_Table_Field_ - Definitions

11.1 Deployments Table Field Definitions

Filename: deployments.csv

Below are the field definitions for the Deployments table, which is a primary table.

Field	Definition	Re- quired	De- fault	Data Type	Refer- ence Table	Con- straint
orga- niza- tion_- code	Code for the organization that owns the deployment Examples: "NEFSC", "WHOI"	Al- ways		Code	Orga- niza- tions (see Excel)	
de- ploy- ment_- code	Unique code for the deployment. The recommendation is to use [organization_code]_[project_region_code]_[YYYYMM]_[site_code].	Al- ways		Code		Unique within organi- zation

Examples:

"WHOI_GOM_202310_WE03",
 "WHOI_MA-RI_202210_WE16",
 "NEFSC_MA-RI_202110_NS01",
 "NEFSC_MA-RI_202202_COX02"

11 11_Deployments_Table_Field_Definitions

project_code	Code of the project that the deployment belongs to. Only need to create a project code if there are multiple deployments that should be grouped together.	Required for NCEI	Code Projects
	Examples: "WHOI_GOM_202310", "NEFSC_MA-RI_202202"		

12 12_Recordings_Table_Field_Definitions

13 13_Analyses_Table_Field_Definitions

14 14_Detections_Table_Field_Definitions

15 15_Devices_Table_Field_Definitions

16 16_Projects_Table_Field_Definitions

17 17_Sites_Table_Field_Definitions

18 18_Recording_Intervals_Table_Field_- Definitions

19 19_Tracks_Table_Field_Definitions

20 20_Track_Positions_Table_Field_ Definitions

21 21_Sensor_Dataset_Field_Definitions

22 22__Sensor__Values__Table__Field__- Definitions

23 23_Frequently_Asked_Questions

24 24_Example_Scenarios